

Full-result packet for arXiv:1605.06909

A contractible, unitarily representable SIN Polish group
that is not of finite type

Status. full_solution_likely_valid. This packet answers Question 4.5 of Hiroshi Ando, Yasumichi Matsuzawa, Andreas Thom, and Asger Törnquist, *Unitarizability, Maurey–Nikishin factorization, and Polish groups of finite type*, arXiv:1605.06909v3.

Source question. A Polish group is *unitarily representable* if it is topologically isomorphic to a subgroup of the unitary group of a separable Hilbert space, with the strong operator topology. It is *of finite type* if it is topologically isomorphic to a closed subgroup of the unitary group of a finite von Neumann algebra with separable predual. The source constructs unitarily representable SIN Polish groups which are not of finite type, but its construction has a discrete quotient. It then asks:

Is there a connected, unitarily representable SIN Polish group which is not of finite type?

We give an affirmative answer with “connected” strengthened to “contractible.” The construction is a general permanence device.

The measurable-function group. Let G be a Polish group and let

$$L^0(G) = L^0([0, 1], G)$$

be the group of equivalence classes, modulo equality almost everywhere, of Lebesgue-measurable maps $f : [0, 1] \rightarrow G$, with pointwise multiplication and the topology of convergence in measure. If $d \leq 1$ is a compatible metric on G , that topology is induced by

$$D_d(f, h) = \int_0^1 d(f(x), h(x)) dx.$$

It is standard, and also follows by approximating with simple functions, that $L^0(G)$ is Polish when G is Polish.

Theorem 1 (The L^0 -permanence device). *Let G be a unitarily representable SIN Polish group. Then $L^0(G)$ is a contractible, unitarily representable SIN Polish group. Moreover, if G is not of finite type, then $L^0(G)$ is not of finite type.*

Proof. We prove the four assertions separately.

SIN and Polishness. Choose a bounded compatible two-sided invariant metric d on G . The integral metric D_d above is two-sided invariant:

$$D_d(af, ah) = D_d(f, h) = D_d(fa, ha) \quad (a, f, h \in L^0(G)).$$

It induces convergence in measure. Thus $L^0(G)$ is SIN, and the standard simple-function completeness and separability argument makes it Polish.

Contractibility. For $0 \leq s \leq 1$, define

$$[C_s f](x) = \begin{cases} f(x), & 0 \leq x \leq 1 - s, \\ e, & 1 - s < x \leq 1, \end{cases}$$

where e is the identity of G . Then $C_0f = f$ and $C_1f = e$. For all $s, t \in [0, 1]$ and $f, h \in L^0(G)$, boundedness of d gives

$$D_d(C_sf, C_th) \leq D_d(f, h) + |s - t|.$$

Hence $(s, f) \mapsto C_sf$ is a continuous contraction of $L^0(G)$ to its identity. In particular, $L^0(G)$ is path connected and connected.

Unitary representability. Let

$$\rho : G \longrightarrow \mathcal{U}(H)$$

be a topological group embedding, where H is separable and $\mathcal{U}(H)$ has the strong operator topology. On $K = L^2([0, 1], H)$, define the multiplication representation

$$[\Pi(f)\xi](x) = \rho(f(x))\xi(x).$$

This is a well-defined injective homomorphism $L^0(G) \rightarrow \mathcal{U}(K)$. We check that it is a topological embedding, which is the only point requiring care.

Choose a dense sequence $(v_j)_{j \geq 1}$ in the unit sphere of H . The bounded metric

$$r(g, h) = \sum_{j=1}^{\infty} 2^{-j} \min\{1, \|\rho(g) - \rho(h)\|v_j\}$$

induces the topology of G . Consequently D_r induces convergence in measure on $L^0(G)$.

Suppose $D_r(f_n, f) \rightarrow 0$. For every j , the bounded functions

$$x \mapsto \|\rho(f_n(x)) - \rho(f(x))\|v_j\|$$

converge to zero in measure, hence in L^2 . Thus $\Pi(f_n) \rightarrow \Pi(f)$ strongly on every constant vector field v_j . Finite-valued fields with values in the span of the v_j 's are dense in K , and all the operators are unitary, so the convergence is strong on all of K .

Conversely, if $\Pi(f_n) \rightarrow \Pi(f)$ strongly, testing on each constant field v_j gives

$$\int_0^1 \|\rho(f_n(x)) - \rho(f(x))\|v_j\|^2 dx \longrightarrow 0.$$

The defining series for r , first truncated and then estimated by its uniformly small tail, yields $D_r(f_n, f) \rightarrow 0$. Therefore Π is a topological embedding into the unitary group of the separable Hilbert space K .

Failure of finite type. The constant-map embedding

$$\iota : G \longrightarrow L^0(G), \quad \iota(g)(x) = g,$$

is a topological group embedding with closed image. To see closedness, if a sequence of constant maps converges in measure, pass to a subsequence which converges almost everywhere. At any two points in the common conull set, the same sequence in G has both pointwise limits, so those limits agree; the limiting function is constant almost everywhere.

Finite type passes to closed subgroups directly from its definition. Thus, if $L^0(G)$ were of finite type, its closed constant subgroup $\iota(G)$ would be of finite type. This proves the contrapositive of the last assertion. $\square$

Corollary 2 (Answer to the source question). *There exists a contractible, unitarily representable SIN Polish group which is not of finite type.*

Proof. Choose a uniformly bounded non-unitarizable representation $\pi : \mathbb{F}_2 \rightarrow \mathrm{GL}(H)$ on a separable Hilbert space and form

$$G_0 = H \rtimes_{\pi} \mathbb{F}_2.$$

The Main Theorem of arXiv:1605.06909 says that G_0 is a unitarily representable SIN Polish group and that it is of finite type if and only if π is unitarizable. Hence G_0 is not of finite type. Apply Theorem 1 to obtain $L^0(G_0)$. $\square$

Checks and scope. The argument uses the source’s definitions of unitary representability and finite type. The multiplication representation is not merely faithful: the constant-vector-field calculation proves that it induces exactly the convergence-in-measure topology. The non-finite-type conclusion does not require a new von Neumann algebra obstruction; it is inherited from the closed subgroup of constant functions. Contractibility is asserted only as a property of the underlying topological space, which is more than the connectedness requested by the source.

Novelty check. Before promotion, the run indexes were searched for arXiv:1605.06909, “connected unitarily representable SIN,” “Polish groups of finite type,” and measurable-function groups; no earlier packet was found. The current arXiv record remains v3 and contains the question. Bounded searches for the exact sentence and for $L^0([0, 1], G)$ together with finite type found the source and general work on measurable-function groups, but no stated answer by this permanence device.

Human-review recommendation. Review as a short full answer. The two points worth checking are the reverse topology implication for the multiplication representation and the use of closed-subgroup heredity for finite type; both are written out above.

References

- [1] H. Ando, Y. Matsuzawa, A. Thom, and A. Törnquist, *Unitarizability, Maurey–Nikishin factorization, and Polish groups of finite type*, arXiv:1605.06909v3, 2017.
- [2] A. Kwiatkowska and M. Malicki, *Automorphism groups of countable structures and groups of measurable functions*, arXiv:1612.03106, 2016.